THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry

Dear Dr. X,

Please consider our paper, entitled "..." for publication as a X in *Nature Plants*.

Understanding how plant reproduction copes with fluctuating climatic conditions has become increasingly important with climate change. Warming could disrupt how plant populations regenerate, with potential impacts on ecosystem functioning, including carbon storage.

Forest ecosystems, in particular, are expected to be especially vulnerable to climate change. In many forests, regeneration success depends on reproductive cycles that are synchronous across most individuals—a phenomenon refered to as masting. Several recent works suggest that rapid shifts in climate could break down this synchrony, with major implications for species adaptation and forest health. Yet, how climate interact with the reproductive biology of individual trees remains unclear, limiting our ability to make mechanistically grounded forecasts.

Building on decades of research on tree reproductive biology and long-term seed production data, we develop an approach that links individual-tree reproduction to population-scale dynamics. In particular, we integrate a well-established reproductive constraint, the temporal overlap between fruit maturation and floral bud initiation, with climatic drivers of masting. [...]

We show that reproductive constraints buffer against the runaway effects of climate change. Even under major warming, trees cannot remain locked in a high reproductive state, preventing major shifts in the frequency of high seed-production years. At the population scale, the interplay between individual constraints and shared climatic cues provides a simple physiological mechanism for synchrony. Cold summers act as the critical hinge, driving most trees into a low reproductive state, and [setting the stage] for a synchronous mast year when a warm summer follows.

Our results bring a new angle to the debate over climate change and masting. Contrary to the expectation that warming is already disrupting masting, we find that synchrony has remained stable over four decades of significant warming. However, synchrony might decline with further warming, because the cold summers that synchronize trees could become increasingly rare.

Our work illustrates a principle that extends beyond forests and masting. Plant biological constraints can fundamentally limit the runaway amplification of climate effects on emergent phenomena at larger scales. Developing a mechanistic understanding of the biological and climatological drivers of plant timing will be critical to forecast regeneration and to understand the long-term adaptive responses of plant communities more broadly.

All authors contributed to this work and approve this version for submission. The manuscript is X words, with X references and X figures, and is not under consideration elsewhere. We hope you find it suitable for publication in *Nature Plants*, and look forward to hearing from you.

Sincerely,

V. Van der Meersch

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References